Short Note on SSP

(Is this a BwK Problem?) A: Yes.

Under a causal policy $\pi \in \Pi_A$,

decision at stage $n \longrightarrow (X_n^\pi, C_n^\pi)$

$$X_n^\pi = \sum_a 1\{I_n^\pi = a\} X_{a,n} \quad \text{if } |\mathcal{A}| < \infty.$$

$$N^\pi(B) = \inf\{n \in \mathbb{N} : \sum_{t=1}^{n} X_t^\pi > B\}$$ stopping time (FPT, FHT)

$$V_B^*(s) = \min_{\pi \in \Pi_A} \mathbb{E}\left[\sum_{n=1}^{N^\pi(B)} C_n^\pi \,\middle|\, \vec{S}_0^\pi = \vec{s}\right] \quad (\text{SSP})$$

States

$$\Sigma_n^\pi = \sum_{t=1}^{n} X_t^\pi \quad (\text{w.l.o.g., } \Sigma_0^\pi = 0)$$

$$T_{a,n}^\pi = \sum_{t=1}^{n} 1\{I_t^\pi = a\}$$

$$\vec{T}_n^\pi = (T_{a,n}^\pi : a \in \mathcal{A})$$

Thus, $\vec{S}_n^\pi = (\Sigma_n^\pi, \vec{T}_n^\pi)$ is the state

State transitions

Choose action $a \longrightarrow T_{a,n}^\pi := T_{a,n-1}^\pi + 1$

$$\Sigma_n^\pi := \Sigma_{n-1}^\pi + 1\{I_{n-1}^\pi = a\} X_{n,a}$$

Our results

$E[X_{n,a}] > 0, \ \forall a$

$E[|X_{n,a}|^p] < \infty$ for $p > 2$

$E[|C_{n,a}|^2] < \infty$

$$\Rightarrow V_B^\pi(s) - V_B^*(s) = O(\log B)$$

Optimal regret if $(X_{a,n}, C_{a,n}) \sim \mathcal{N}(\mu_a, \Sigma_a)$

<u>Simplistic model</u> : $|\mathcal{X}| < \infty$

$$(X_{a,n}, C_{a,n}) \overset{iid}{\sim} F_a, \ \forall a \in \mathcal{X}$$

Continuous $\mathcal{S}$ (since $X_{n,a}$ is random)

Yet it can be important.

1. Markovian transitions : rested $\rightarrow$ easy
   restless $\rightarrow$ hard
2. Continuous $\mathcal{X}$: straightforward

☆ 3. Any discounted cost MDP can be translated into an SSP [(Bertsekas, Tsitsiklis ; 1996) Section 2.3]

4. Renewal / Regenerative Processes
   Martingale Concentration
   Lyapunov theory (for constrained opt)
   } strong analysis tools.

5. Connection to other RL problems ?
   (e.g., Q-Learning)